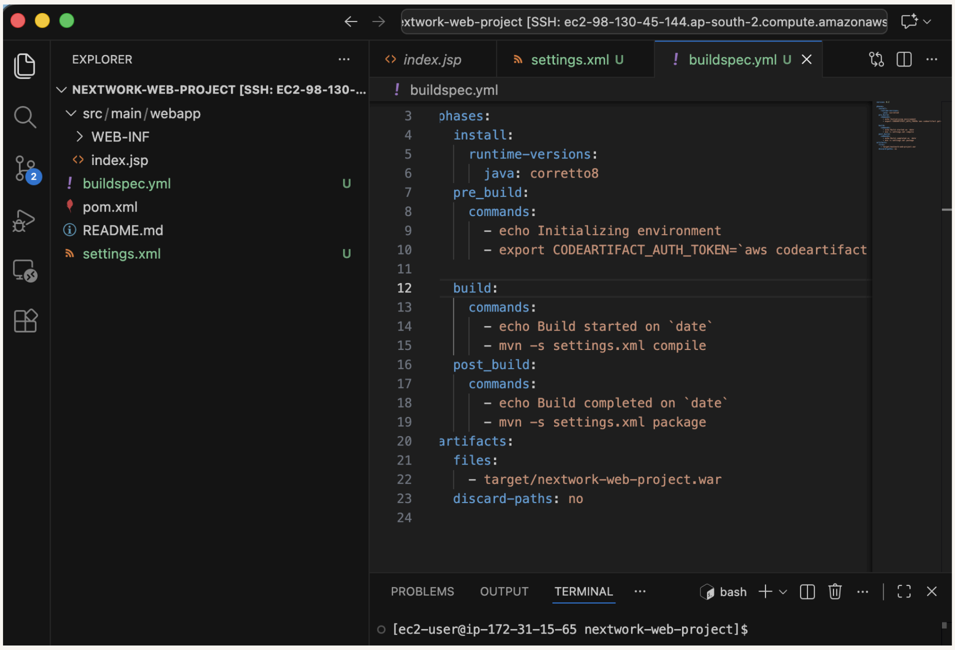
nextwork.org

Continuous Integration with CodeBuild

sahini sourya



Introducing Today's Project!

In this project, I will demonstrate how to use AWS CodeBuild to automate the build process in the CI/CD pipeline. Building means compressing my web application file to single compressed file(like zip file) that i can deploy later. I'm doing this project to learn on codebuild and creating that file automatically.

Key tools and concepts

Services I used were CodeBuild, CodeArtifact, CodeConnection, S3, EC2, GitHub, VS code. Key concepts I learnt include build process, resolving git issues in terminal, buildspec.yml file, and also using CodeConnections to connect AWS with GitHub.

Project reflection

This project took me approximately 3 hours to do including troubleshooting time. The most challenging part was resolving the Git permission issues to make for GitHub repository code changes, Adding permission to both CodeArtifact and CodeBuild as it both needs to acess maven package and settings.yml file. It was most rewarding to see the CodeBuild succeed.

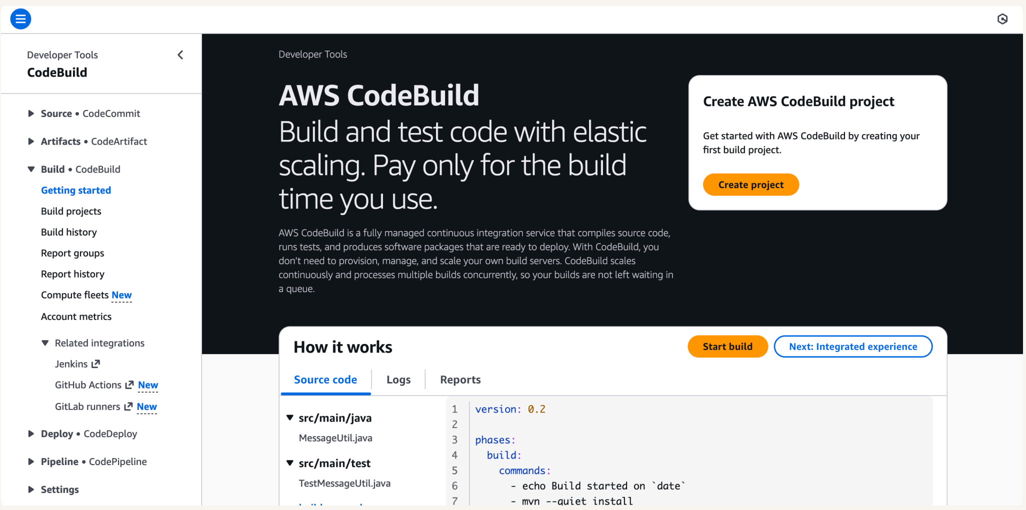
This project is part four of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow which is 5th day of 7 day series.

sahini sourya
NextWork Student

Setting up a CodeBuild Project

CodeBuild is a continuous integration service, which means it helps me to compile and package code (i.e,. turn web application code into something computers can run, then compress it all into neat package). Engineering teams use it because CI services automate this process for us. If it doesn't exist engineers have to compile and package everything manually themselves before deploying the application.

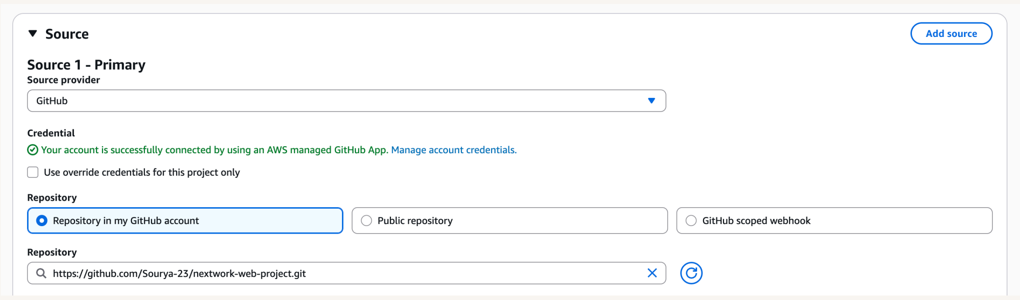
My CodeBuild project's source configuration means it defines where my application source code is located, it tells AWS codebuild where to find the code it needs to build, test, and package. I selected source as github because it is where my web application source code is located and telling as it is to AWS codebuild.



Connecting CodeBuild with GitHub

There are multiple credential types for GitHub, like Github App, Personal access token and OAuth app. I used GitHub App because it is generally most simplest and most secure option.

The service that helped connect my AWS environment with GitHub is CodeConnections. CodeConnections also lets us connect my environement to other third party platforms, in this case it make sure i have a secure connection to gihub so that codebuild project can use our gihub repository and its source with permission.



CodeBuild Configurations

Environment

My CodeBuild project's Environment configuration means the environment setup for compute power i.e, the EC2 instance that will be compiling and compressing the files. when i run my CodeBuild project. It includes settings like provisioning model compute operating system, image and service role. These settings determine how my new EC2 instance will be spun up and conduct the build process. Its important here that the EC2 instance has the correct permissions to everything it needs to work properly.

Artifacts

Build artifacts are files/resources that get created as a part of the CodeBuild build process. They're important because they reprsent Artifacts that needs to be used later on, in cicd pipeline. For example my build process will create a compressed file that represents my web application and i'll need to use the compressed file to deploy the web app later on. To store my BuildArtifacts, I created an S3 bucket.

Packaging

When setting up CodeBuild, I also chose to package artifacts in a Zip file because this helps with package management as files are Organized neatly in a single excutable file. It also helps me to compress the size of my compiled web application!

Monitoring

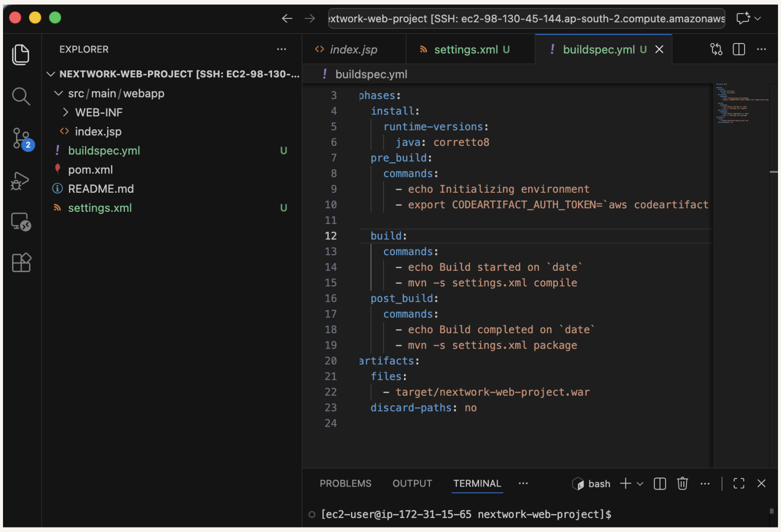
For monitoring, I enabled CloudWatch Logs, which is a service that will note down the commands that i run if any errors are happen. This is so helpful for troubleshooting!

sahini sourya
NextWork Student

buildspec.yml

My first build failed because it doesn't have a buildspec.yml file in the source code. A buildspec.yml file is needed because the CodeBuild reads this file like a step-by-step intruction manual through phases like install, pre_build, build, and then post_build. The beauty of using buildspec.yml is that my build process is defined as code right alongside my web application. This means my build process can be versioned, reviewed, and evolved just like any other part of your codebase.

The first two phases in my buildspec.yml file installs Java and gets us access to CodeArtifact. The third phase in my buildspec.yml file compiles the web application. The fourth phase in my buildspec.yml file packages the web appliaction in single compressed artifact.

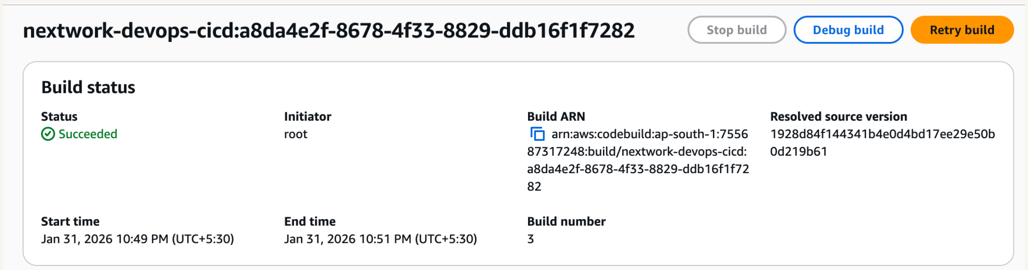


Success!

My second build also failed, but with a different error that said error while excuting mvn -s settings.xml compile, which means my CodeBuild is unable to compile settings.xml, cause of this reason is that CodeBuild doesn't have permission to read CodeArtifact in CodeBuild service role . To fix this i should add CodeArtifact access permission.

To resolve the second error, I simply can add permission of my customized permission named "codeartifact-nextwork-consumer-policy", which will given access to read maven packages as given to CodeArtifact. When I built my project again, I saw successful build conformation this time.

To verify the build, I checked my artifact bucket in S3. Seeing the artifact tells me CodeBuild was success! The build process compilied the code from source repository[GitHub] compressed and stored that file in S3 bucket.



nextwork-devops-cicd:a8da4e2f-8678-4f33-8829-ddb16f1f7282
Stop build
Debug build
Retry build
Build status
Status
Succeeded
Initiator
root
Build ARN
arn:aws:codebuild:ap-south-1:755687317248:build/nextwork-devops-cicd:a8da4e2f-8678-4f33-8829-ddb16f1f7282
Resolved source version
1928d84f144341b4e0d4bd17ee29e50b0d219b61
Start time
Jan 31, 2026 10:49 PM (UTC+5:30)
End time
Jan 31, 2026 10:51 PM (UTC+5:30)
Build number
3

Automating Testing

In a project extension, I'm I added a simple test script that validates the tests on my project structure - is there a src folder and an index.jsp file in your project directory or not?

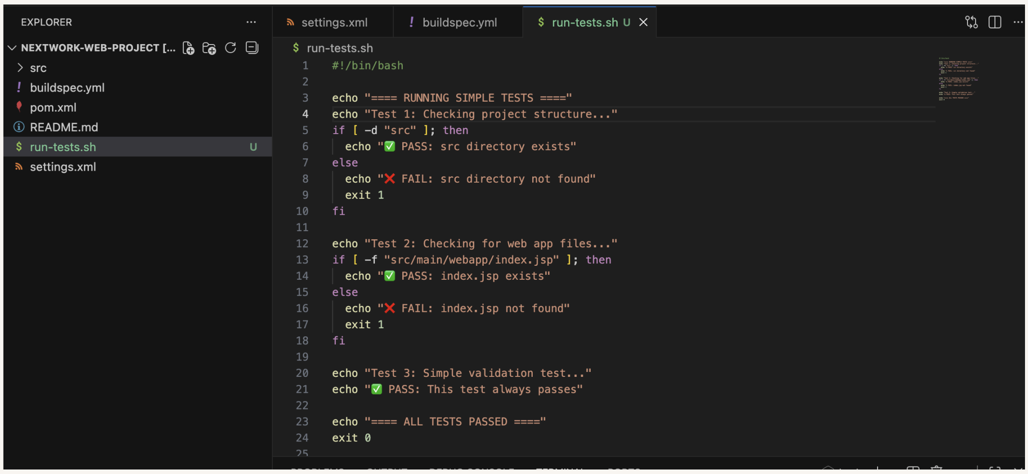
To add the test script to the build process, I updated buildspec.yml file in build commands in which i have now added test script that will work in now CodeBuild as well!!

After pushing my code to GitHub, I ran again to restart the CodeBuild to check the updated version (i.e., added test script in build commands). I could see my CodeBuild is success as when i checked by running test script in locally but now in CodeBuild as well, this imply that this test also automated testing in CodeBuild.

sahini sourya

NextWork Student

nextwork.org



The place to learn & showcase your skills

Check out nextwork.org for more projects